



Cambridge International AS & A Level

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FURTHER MATHEMATICS

9231/33

Paper 3 Further Mechanics

May/June 2025

1 hour 30 minutes

You must answer on the question paper.

You will need: List of formulae (MF19)

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.
- Where a numerical value for the acceleration due to gravity (g) is needed, use 10 m s^{-2} .

INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

This document has **16** pages. Any blank pages are indicated.



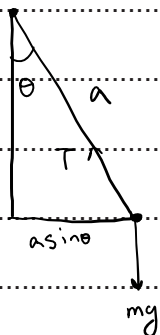
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- 1 A particle P of mass m is attached to one end of a light inextensible string of length a . The other end of the string is attached to a fixed point O . The particle moves in a horizontal circle with constant angular speed ω and with the string inclined at an angle of θ to the downward vertical.

Given that $\tan \theta = \frac{4}{3}$, find ω in terms of a and g .

[3]



$$T \cos \theta = mg$$

$$T = \frac{5}{3} mg$$

$$\tan \theta = \frac{4}{3}$$

$$\sin \theta = \frac{4}{5}$$

$$\cos \theta = \frac{3}{5}$$

$$T \sin \theta = m r \omega^2$$

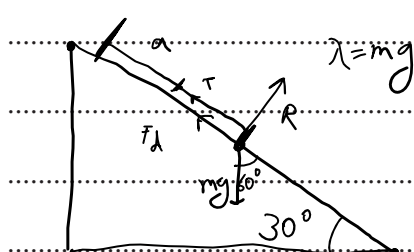
$$\frac{5}{3} mg \sin \theta = m a \sin \theta \omega^2$$

$$\omega = \sqrt{\frac{5g}{3a}} \text{ rad s}^{-1}$$



- 2 A particle P of mass m is attached to one end of a light elastic string of natural length a and modulus of elasticity mg . The other end of the string is attached to a fixed point O on a rough plane inclined at an angle of 30° to the horizontal. The particle P is held at rest at point O before being released. The frictional force acting on P as it slides down the plane is $\frac{11}{30}mg$.

(a) Find, in terms of a , the distance that P moves down the plane before coming to rest. [5]



→ Lets say it slides a distance ' ka ' totally

Initially:

$$PE = 0$$

$$KE = 0$$

$$EPE = 0$$

Finally:

$$KE = 0$$

$$PE = (-ka \sin 30)(mg)$$

$$EPE = \frac{(mg)(ka-a)^2}{2a}$$

$$\text{Work against friction} = \frac{11}{30} mg ka$$

CoE:

$$-\frac{mgak}{2} + \frac{mg a^2 (k-1)^2}{2a} + \frac{11}{30} mg ka = 0$$

$$-\frac{k}{2} + \frac{k^2 - 2k + 1}{2} + \frac{11}{30} k = 0$$

$$-15k + 15k^2 - 30k + 15 + 11k = 0$$

$$15k^2 - 34k + 15 = 0$$

$$15k^2 - 25k - 9k + 15 = 0$$

$$5k(3k-5) - 3(3k-5) = 0$$

$$k = \frac{5}{3} \text{ or } \frac{3}{5}$$

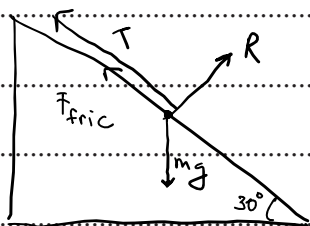
↳ $k > 1$ so $k = \frac{5}{3}$ → slides $\frac{5}{3}a$ m



(b) It is given that P remains at rest in this new position.

Find, in terms of m and g , the magnitude of the frictional force in this position.

[3]



$$mg \cos 30 = R$$

$$\frac{\sqrt{3}}{2} mg = R$$

$$\mu R + \frac{\lambda}{\lambda} = mg \sin 30$$

$$\mu \left(\frac{\sqrt{3}}{2} mg \right) + \frac{mg \left(\frac{2}{3} \right)}{2} = \frac{mg}{2}$$

$$\mu \left(\frac{\sqrt{3}}{2} \right) = \frac{1}{2} - \frac{2}{3}$$

$$\mu = \left(\frac{-1}{6} \right) \left(\frac{2}{\sqrt{3}} \right) = \frac{-2}{6\sqrt{3}}$$

↳ note that $\mu < 0$ as F_{fric} acts in the other direction

$$\rightarrow \mu R = \left(\frac{-2}{6\sqrt{3}} \right) \left(\frac{\sqrt{3}}{2} mg \right) = \boxed{\frac{1}{6} mg}$$





- 3 A ball of mass m kg is projected vertically upwards with initial speed U m s⁻¹ and moves under gravity. At time t s after projection, the ball has travelled a distance x m and its speed is v m s⁻¹. There is a resistive force of magnitude mkv^2 N, where k is a positive constant.

- (a) Show that the distance travelled by the ball when it is moving upwards is $x = \frac{1}{2k} \ln \left(\frac{g+kU^2}{g+kv^2} \right)$. [4]

$$F_{\text{down}} = mg + mkv^2 \Rightarrow a = g + kv^2$$

$$v \frac{dv}{dx} = g + kv^2$$

$$\int \frac{v}{g+kv^2} dv = \int dx$$

$$\begin{aligned} \text{let } n &= g + kv^2 \\ dn &= 2kv \, dv \end{aligned}$$

$$x + C = \int \left(\frac{v}{n} \right) \frac{1}{2kv} dn$$

$$x + C = \frac{1}{2k} \ln |n|$$

$$x + C = \frac{1}{2k} \ln |g + kv^2| \quad \rightarrow \text{when } x=0, v=U$$

$$C = \frac{1}{2k} \ln |g + kU^2|$$

$$x = \frac{1}{2k} \ln |g + kv^2| - \frac{1}{2k} \ln |g + kU^2|$$

$$x = \frac{1}{2k} \ln \left| \frac{g + kv^2}{g + kU^2} \right|$$

downward

$$x = \frac{-1}{2k} \ln \left(\frac{g + kU^2}{g + kv^2} \right) = \frac{1}{2k} \left(\ln \left(\frac{g + kv^2}{g + kU^2} \right) \right)$$





It is given that $k = 0.025$ and that $U = 20$.

- (b) Find the time taken for the ball to reach its maximum height.

[4]

$$a = g + 0.025v^2 \text{ down}$$

$$\frac{dv}{dt} = -(g + 0.025v^2)$$

$$\frac{dv}{dt} = \frac{-1}{40} (v^2 + 400)$$

$$\int \frac{1}{v^2 + 20^2} dv = \int \frac{-1}{40} dt$$

$$\frac{1}{20} \tan^{-1}\left(\frac{v}{20}\right) = \frac{-1}{40} t + C$$

$$C = \frac{1}{20} \tan^{-1}(1) = \frac{\pi}{80}$$

$$\tan^{-1}\left(\frac{v}{20}\right) = -\frac{1}{2}t + \frac{\pi}{4}$$

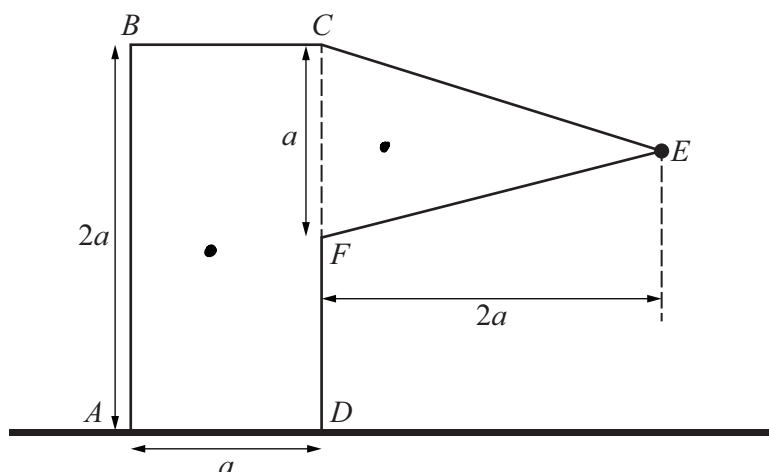
$$v = 20 \tan\left(-\frac{1}{2}t + \frac{\pi}{4}\right)$$

$$\text{@ max height, } v=0$$

$$\tan\left(-\frac{1}{2}t + \frac{\pi}{4}\right) = 0$$

$$t = \frac{\pi}{2} \text{ s}$$





An object consists of a uniform lamina with a particle attached. The uniform lamina $ABCEFD$ of mass m is formed from a rectangle $ABCD$ and an isosceles triangle CEF , where F is the midpoint of CD . The rectangle has sides $AB = 2a$ and $AD = a$. The triangle CEF has base a and height $2a$. The particle of mass km is attached to the lamina at E . The object rests in a vertical plane with its edge AD on horizontal ground (see diagram).

Given that the object is on the point of toppling in its vertical plane about the vertex D , find the value of k . [4]

(with D: (0, 0))	ABCD	CEF	Particle
mass	$2a^2\rho$	$a^2\rho$	km
COM	$(-a/2, a)$	$(2a/3, 3a/2)$	$(2a, 3a/2)$

$$\hookrightarrow 3a^2\rho = m \Rightarrow \rho = \frac{1}{3a^2}$$

$\rightarrow$ If on point of toppling, $\bar{x} = 0$

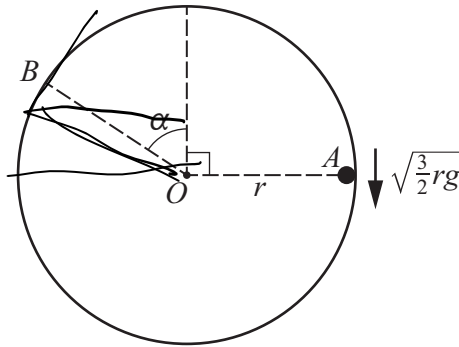
$$\bar{x}(m+km) = \left(\frac{2}{3}m\right)\left(-\frac{a}{2}\right) + \left(\frac{m}{3}\right)\left(\frac{2a}{3}\right) + km(2a)$$

$$0 = \frac{-am}{3} + \frac{2am}{9} + 2kma$$

$$2k = 1/9 \Rightarrow k = 1/18$$



[illegible]



A hollow cylinder of radius r is fixed with its axis horizontal. Points A , B and O are in the same vertical plane perpendicular to the axis of the cylinder, with A and B on the smooth inner surface and O on the axis. OA and OB make angles 90° and α respectively with the upward vertical through O , with A and B on opposite sides of the vertical. A particle of mass m is projected vertically downwards from point A with speed $\sqrt{\frac{3}{2}rg}$ and moves in a vertical circle inside the cylinder (see diagram). The particle loses contact with the cylinder at point B .

(a) Find the value of α .

[4]

Let speed @ bottom = n

$$\text{@ } A: \frac{mv^2}{r} = \frac{mn^2}{r} - 2mg + 2mg \cos 90^\circ$$

$$\frac{m\left(\frac{3}{2}rg\right)}{r} + 2mg = \frac{mn^2}{r}$$

$$\sqrt{\frac{7}{2}rg} = n$$

@ B, $R=0$

$$R = \frac{mn^2}{r} - 2mg + 3mg \cos (180 - \alpha)$$

$$0 = m\left(\frac{7/2rg}{r} - 2g + 3g \cos (180 - \alpha)\right)$$

$$\frac{3}{2}g = 3g \cos (180 - \alpha)$$

$$\cos \alpha = \frac{1}{2} \Rightarrow \alpha = 60^\circ$$



- (b) In the subsequent motion find, in terms of r , the greatest height above O reached by the particle. [4]

@ B:

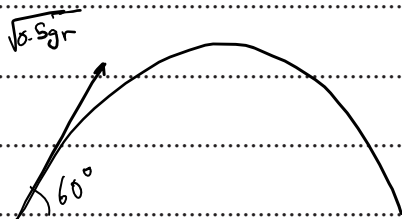
$$\frac{mv^2}{r} = \frac{mn^2}{r} - 2mg + 2mg \cos(180 - \alpha)$$

$$v^2 = \left(\sqrt{\frac{7}{2}gr}\right)^2 - 2gr - 2gr \cos(60)$$

$$v^2 = \frac{7}{2}gr - 2gr - gr = \frac{1}{2}gr$$

$$\hookrightarrow v = \sqrt{0.5gr}$$

Consider the projectile:



$$v_y = u \sin \theta - gt \rightarrow @ \text{ max } H, v_y = 0$$

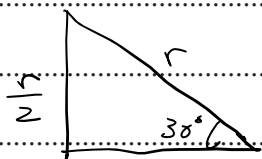
$$\hookrightarrow t = \frac{u \sin \theta}{g}$$

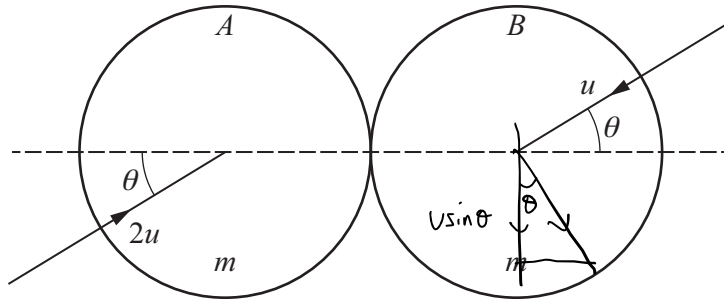
$$s = (u \sin \theta) \left(\frac{u \sin \theta}{g} \right) - \frac{1}{2} g \left(\frac{u \sin \theta}{g} \right)^2$$

$$= \frac{u^2 \sin^2 \theta}{2g} = \frac{(0.5gr) \left(\frac{3}{4} \right)}{2g}$$

$$= \frac{3}{16}r$$

$$\rightarrow \text{Max height} = \frac{r}{2} + \frac{3}{16}r = \frac{11}{16}r$$





Two identical uniform smooth spheres A and B , each with mass m , are moving on a horizontal surface with speeds $2u$ and u respectively when they collide. Immediately before the collision, the spheres are moving parallel to each other in opposite directions such that their directions of motion each make an angle θ with the line of centres (see diagram). As a result of the collision, B moves in a direction which is perpendicular to its initial direction of motion. The coefficient of restitution between the spheres is e .

- (a) Find an expression for $\tan \theta$ in terms of e .

[6]

CoM:

$$m(2u \cos \theta) - m(u \cos \theta) = mv_{xA} + mv_{xB}$$

$$u \cos \theta = v_{xA} + v_{xB} \quad (\text{right as +ve})$$

Newton's:

$$\frac{v_{xB} - v_{xA}}{3u \cos \theta} = e$$

$$3e u \cos \theta = (v_{xB} - v_{xA})$$

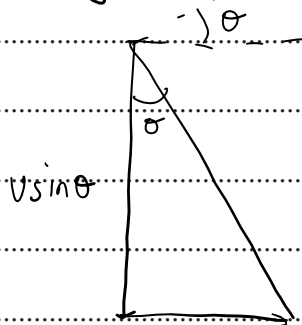
$$3eu \cos \theta = v_{xB} - v_{xA}$$

$$+ \quad u \cos \theta = v_{xB} + v_{xA}$$

$$v_{xB} = u \cos \theta \left(\frac{3e+1}{2} \right)$$



Analyzing B:



$$\tan \theta = \frac{V_B}{u \sin \theta}$$

$$\tan \theta = \frac{u \cos \theta \left(\frac{3e+1}{2} \right)}{u \sin \theta}$$

$$\tan^2 \theta = \frac{3e+1}{2}$$

$$\tan \theta = \sqrt{\frac{3e+1}{2}}$$

As a result of the collision, *A* moves in a direction which is perpendicular to the line of centres.

(b) Find the value of θ .

[2]

$$V_{2A} = 0$$

$$\hookrightarrow V_{2B} = 3eu \cos \theta = u \cos \theta$$

$$\hookrightarrow e = \frac{1}{3}$$

$$\tan \theta = 1$$

$$\hookrightarrow \theta = \pi/4$$



- 7 A particle P is projected from a point O with speed U at an angle 45° above the horizontal and moves freely under gravity.

(a) State the vertical and horizontal components of velocity at time t . [1]

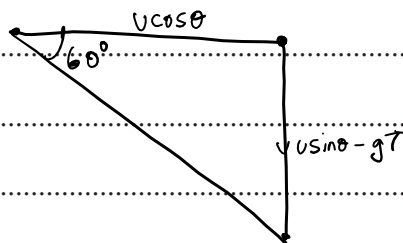
$$V_y = U \sin \theta - gt$$

$$V_x = U \cos \theta$$

At time T , particle P is moving at an angle of 60° below the horizontal.

(b) Show that $T = \frac{U}{2g}(\sqrt{2} + \sqrt{6})$. [3]

$$\therefore \text{ @ } t = T, \quad \tan(-60^\circ) = \frac{U \sin \theta - gT}{U \cos \theta}$$



$$-\sqrt{3} = \tan \theta - \frac{gT}{U \cos \theta}$$

$$-\sqrt{3} = 1 - \frac{gT}{U(\frac{1}{\sqrt{2}})}$$

$$\frac{gT\sqrt{2}}{U} = 1 + \sqrt{3}$$

$$T = \frac{U}{g} \left(\frac{1}{\sqrt{2}} + \frac{\sqrt{3}}{\sqrt{2}} \right)$$

$$T = \frac{U}{2g} (\sqrt{2} + \sqrt{6})$$



At time T , the particle strikes a smooth horizontal plane at a point which is a horizontal distance D from O and a vertical distance H below O .

(c) Find the ratio $H : D$.

[4]

@ Time t : $D = v t \cos \theta = \frac{v^2}{2g} (\sqrt{2} + \sqrt{6}) \cdot \left(\frac{1}{\sqrt{2}}\right) = \frac{v^2}{2g} (1 + \sqrt{3})$

$$y = x \tan(45) - \frac{g x^2}{2v^2 \sec^2 45}$$

$$y = x - \frac{g x^2}{v^2}$$

$$H = \frac{v^2}{2g} (1 + \sqrt{3}) - \frac{g}{v^2} \left(\frac{v^2}{2g} (1 + \sqrt{3}) \right)^2$$

$$= \frac{v^2}{2g} (1 + \sqrt{3}) - \frac{v^2}{4g} (4 + 2\sqrt{3})$$

$$= \frac{v^2}{2g} (H\sqrt{3} - 2 - \sqrt{3}) = \frac{-v^2}{2g}$$

$$\hookrightarrow H : D =$$

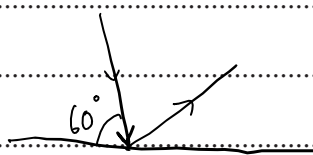
$$\frac{v^2}{2g} : \frac{v^2}{2g} (1 + \sqrt{3})$$

$$\hookrightarrow \boxed{1 : 1 + \sqrt{3}}$$

After striking the horizontal plane, P rebounds with speed w . The coefficient of restitution between P and the plane is $\frac{2}{3}$.

(d) Find w in terms of U .

[3]



Initially:

$$U_x = \frac{U}{\sqrt{2}}$$

$$U_y = \frac{U}{\sqrt{2}} - \frac{U}{2} (\sqrt{2} + \sqrt{6})$$

$$= \frac{U\sqrt{6}}{2} \text{ downward}$$

Finally:

$$v_x = \frac{U}{\sqrt{2}}$$

$$v_y = \frac{2}{3} \left(\frac{U\sqrt{6}}{2} \right) = \frac{U\sqrt{6}}{3}$$

$$w = \sqrt{\left(\frac{U}{\sqrt{2}} \right)^2 + \left(\frac{U\sqrt{6}}{3} \right)^2}$$

$$= \sqrt{\frac{U^2}{2} + \frac{2U^2}{3}} = \boxed{U\sqrt{\frac{7}{6}}}$$



[illegible]

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